

# pacemaker

Test Specification

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## Introduction

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This document contains test cases for the **pacemaker** library. Each section maps one-to-one to a requirement section in the Requirements Specification document. Test cases are expressed in Given-When-Then (GWT) notation and supplemented with explicit step tables for automation engineers.

### Test strategy:

- Because the library wraps a Windows-only COM server (INCA), all integration tests require a Windows host with INCA installed.
- Unit tests for classes that do not make live COM calls (e.g. `unique_com_ptr`, `COMProxy`) shall use mock COM objects implemented as in-process test doubles.
- Tests that cannot run without INCA shall be tagged `[integration]`.
- Tests runnable with mocks shall be tagged `[unit]`.

**TC-001 – unique\_com\_ptr**

Requirement reference: REQ-001.

**TC-001-A: Release is called on destruction**

**Tags:**

[unit]

**Requirement:**

AC-001-1

**Given** a mock COM object whose **Release()** method increments a call counter.

**When** a `unique_com_ptr` owning that object goes out of scope.

**Then** **Release()** has been called exactly once and the counter equals 1.

| # | Step                                                              | Expected result                               |
|---|-------------------------------------------------------------------|-----------------------------------------------|
| 1 | Create a mock COM object with a release counter initialised to 0. | Object is alive; counter = 0.                 |
| 2 | Wrap it in <code>unique_com_ptr&lt;MockCOM&gt;</code> .           | Wrapper holds the pointer; counter still 0.   |
| 3 | Let the wrapper go out of scope (end of block).                   | Destructor fires; <b>Release()</b> is called. |
| 4 | Assert <code>counter == 1</code> .                                | Test passes.                                  |

**TC-001-B: No Release on null pointer**

**Tags:**

[unit]

**Requirement:**

AC-001-2

**Given** a default-constructed (null) `unique_com_ptr<MockCOM>`.

**When** the wrapper is destroyed.

**Then** no crash occurs and the deleter does not dereference the null pointer.

| # | Step                                                                                   | Expected result                   |
|---|----------------------------------------------------------------------------------------|-----------------------------------|
| 1 | Default-construct <code>unique_com_ptr&lt;MockCOM&gt;</code> <code>p{nullptr}</code> . | <code>p.get() == nullptr</code> . |
| 2 | Destroy <code>p</code> .                                                               | No crash, no undefined behaviour. |

**TC-001-C: Move transfers ownership without extra Release**

**Tags:**

[unit]

**Requirement:**

AC-001-3

**Given** a `unique_com_ptr` owning a mock COM object (release counter = 0).

**When** the pointer is moved into a second `unique_com_ptr` and both go out of scope.

**Then** `Release()` is called exactly once and the moved-from wrapper is null.

| # | Step                                                         | Expected result                                                                  |
|---|--------------------------------------------------------------|----------------------------------------------------------------------------------|
| 1 | Wrap mock in <code>unique_com_ptr a</code> .                 | <code>a.get() != nullptr</code> ; counter = 0.                                   |
| 2 | Move-construct <code>unique_com_ptr b{std::move(a)}</code> . | <code>a.get() == nullptr</code> ; <code>b.get() != nullptr</code> ; counter = 0. |
| 3 | Destroy <code>b</code> .                                     | <code>Release()</code> called once; counter = 1.                                 |
| 4 | Destroy <code>a</code> (now null).                           | No additional <code>Release()</code> .                                           |

**TC-001-D: Copy construction is deleted****Tags:**

[unit] (compile-time check)

**Requirement:**

AC-001-4

**Given** a `unique_com_ptr<T>` type.

**When** a copy-construction expression is compiled.

**Then** the compilation fails with a deleted-function error.

| # | Step                                                                                                   | Expected result                                   |
|---|--------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| 1 | Write <code>static_assert(!std::is_copy_constructible_v&lt;unique_com_ptr&lt;MockCOM&gt;&gt;)</code> . | Assertion holds; test file compiles successfully. |

**TC-002 – COMProxy<T>**

Requirement reference: REQ-002.

**TC-002-A: Copy is deleted, move is available****Tags:**

[unit] (compile-time check)

**Requirement:**

AC-002-1, AC-002-2

**Given** a concrete class `ConcreteProxy` that derives from `COMProxy<MockCOM>` and exposes the base constructor.

**When** copy and move constructibility are queried at compile time.

**Then** `std::is_copy_constructible_v<ConcreteProxy>` is false and `std::is_move_constructible_v<ConcreteProxy>` is true.

| # | Step                                                                                               | Expected result         |
|---|----------------------------------------------------------------------------------------------------|-------------------------|
| 1 | Define <code>ConcreteProxy</code> deriving from <code>COMProxy&lt;MockCOM&gt;</code> in test code. | Compiles without error. |
| 2 | Assert <code>!std::is_copy_constructible_v&lt;ConcreteProxy&gt;</code> .                           | Static assertion holds. |
| 3 | Assert <code>std::is_move_constructible_v&lt;ConcreteProxy&gt;</code> .                            | Static assertion holds. |

**TC-002-B: Constructor takes ownership via move****Tags:**

[unit]

**Requirement:**

AC-002-3, AC-002-4

**Given** a mock COM object wrapped in a `unique_com_ptr`.

**When** the pointer is passed to a `COMProxy`-derived constructor.

**Then** the proxy's protected `p_subject` holds the pointer and `Release()` has not yet been called.

| # | Step                                                                 | Expected result                                    |
|---|----------------------------------------------------------------------|----------------------------------------------------|
| 1 | Create mock with counter = 0; wrap in <code>unique_com_ptr</code> .  | Counter = 0.                                       |
| 2 | Construct <code>ConcreteProxy</code> from the pointer (moves it in). | Proxy alive; counter = 0.                          |
| 3 | Destroy proxy.                                                       | Counter = 1 (exactly one <code>Release()</code> ). |

## TC-003 – IncaProxy

Requirement reference: REQ-003.

*Note: tests in this section require either a running INCA instance ([integration]) or a mock COM server ([unit]).*

### TC-003-A: Construction narrows IDispatch successfully

**Tags:**

[unit]

**Requirement:**

AC-003-1

**Given** a mock IDispatch that implements the Inca\_Dispatch interface UUID.

**When** IncaProxy is constructed with that pointer.

**Then** no exception is thrown and the proxy is valid.

| # | Step                                             | Expected result            |
|---|--------------------------------------------------|----------------------------|
| 1 | Create mock implementing Inca_Dispatch UUID.     | Mock ready.                |
| 2 | Construct IncaProxy proxy{std::move(idispatch)}. | No exception; proxy alive. |

### TC-003-B: Destructor calls DisconnectFromTool

**Tags:**

[unit]

**Requirement:**

AC-003-2

**Given** a mock Inca\_Dispatch that records calls to DisconnectFromTool.

**When** the owning IncaProxy is destroyed.

**Then** DisconnectFromTool() has been called exactly once.

| # | Step                              | Expected result                   |
|---|-----------------------------------|-----------------------------------|
| 1 | Create mock; wrap in IncaProxy.   | Call counter = 0.                 |
| 2 | Destroy IncaProxy (end of scope). | DisconnectFromTool() counter = 1. |

### TC-003-C: GetOpenedExperiment returns valid IDispatch

**Tags:**

[unit]

**Requirement:**

AC-003-3

**Given** a mock `Inca_Dispatch` whose `GetOpenedExperiment()` returns a non-null `IDispatch*`.**When** `proxy.GetOpenedExperiment()` is called.**Then** the returned `unique_com_ptr<IDispatch>` is non-null.

| # | Step                                                                  | Expected result                         |
|---|-----------------------------------------------------------------------|-----------------------------------------|
| 1 | Configure mock to return a valid experiment <code>IDispatch*</code> . | Mock ready.                             |
| 2 | Call <code>proxy.GetOpenedExperiment()</code> .                       | Returns non-null pointer; no exception. |

**TC-003-D: GetOpenedExperimentView returns valid IDispatch****Tags:**

[unit]

**Requirement:**

AC-003-4

**Given** a mock `Inca_Dispatch` whose `GetOpenedExperimentView()` returns a non-null `IDispatch*`.**When** `proxy.GetOpenedExperimentView()` is called.**Then** the returned `unique_com_ptr<IDispatch>` is non-null.

| # | Step                                                            | Expected result                         |
|---|-----------------------------------------------------------------|-----------------------------------------|
| 1 | Configure mock to return a valid view <code>IDispatch*</code> . | Mock ready.                             |
| 2 | Call <code>proxy.GetOpenedExperimentView()</code> .             | Returns non-null pointer; no exception. |



## TC-004 – IncaOnlineExperimentProxy

Requirement reference: REQ-004.

### TC-004-A: GetAllDevices returns correct count

**Tags:**

[unit]

**Requirement:**

AC-004-2

**Given** a mock IncaOnlineExperiment\_Dispatch whose GetAllDevices() returns a SAFEARRAY containing two VT\_DISPATCH elements.

**When** proxy.GetAllDevices() is called.

**Then** the returned vector has exactly two elements, each non-null.

| # | Step                                                                    | Expected result      |
|---|-------------------------------------------------------------------------|----------------------|
| 1 | Build a SAFEARRAY of 2 VARIANTS, each VT_DISPATCH with mock IDispatch*. | Array ready.         |
| 2 | Configure mock to return the array from GetAllDevices().                | Mock ready.          |
| 3 | Call proxy.GetAllDevices().                                             | No exception thrown. |
| 4 | Assert result.size() == 2.                                              | Passes.              |
| 5 | Assert result[0] != nullptr && result[1] != nullptr.                    | Passes.              |

### TC-004-B: GetAllDevices throws on wrong SAFEARRAY type

**Tags:**

[unit]

**Requirement:**

AC-004-3

**Given** a mock returning a \_variant\_t of type VT\_I4 (wrong type) from GetAllDevices().

**When** proxy.GetAllDevices() is called.

**Then** std::runtime\_error is thrown.

| # | Step                                                                                              | Expected result                         |
|---|---------------------------------------------------------------------------------------------------|-----------------------------------------|
| 1 | Configure mock to return <code>_variant_t{42L}</code> (VT_I4) from <code>GetAllDevices()</code> . | Mock ready.                             |
| 2 | Call <code>proxy.GetAllDevices()</code> inside <code>EXPECT_THROW</code> .                        | <code>std::runtime_error</code> thrown. |

#### TC-004-C: `GetCalibrationValueInDevice` throws when COM returns null

**Tags:**

[unit]

**Requirement:**

AC-004-7

**Given** a mock `IncaOnlineExperiment_Dispatch` whose `GetCalibrationValueInDevice()` returns a null `IDispatch`.

**When** `proxy.GetCalibrationValueInDevice("unknownParam", device)` is called.

**Then** `std::runtime_error` is thrown with a message containing the parameter name.

| # | Step                                                                          | Expected result                         |
|---|-------------------------------------------------------------------------------|-----------------------------------------|
| 1 | Configure mock to return null from <code>GetCalibrationValueInDevice</code> . | Mock ready.                             |
| 2 | Call the method with <code>name = "unknownParam"</code> .                     | <code>std::runtime_error</code> thrown. |
| 3 | Assert exception message contains <code>"unknownParam"</code> .               | Passes.                                 |

#### TC-004-D: `StartRecording` / `StopRecordingAndSave` / `StopMeasurement` delegate correctly

**Tags:**

[unit]

**Requirement:**

AC-004-4, AC-004-5, AC-004-6

**Given** a mock that records calls to `StartRecording`, `StopRecordingAndSave`, and `StopMeasurement`.

**When** each proxy method is called once.

**Then** each underlying COM method has been called exactly once.

| # | Step                                  | Expected result                     |
|---|---------------------------------------|-------------------------------------|
| 1 | Call<br>proxy.StartRecording().       | Mock StartRecording counter = 1.    |
| 2 | Call<br>proxy.StopRecordingAndSave(). | Mock StopRecordingAndSave counter = |
| 3 | Call<br>proxy.StopMeasurement().      | Mock StopMeasurement counter = 1.   |

## TC-005 – IncaExperimentViewProxy

Requirement reference: REQ-005.

### TC-005-A: OpenViewForExperimentDataItem forwards raw pointer

**Tags:**

[unit]

**Requirement:**

AC-005-2

**Given** a mock `IncaExperimentView_Dispatch` that records the `IDispatch*` argument passed to `OpenViewForExperimentDataItem`; and a separate mock data-item `IDispatch`.

**When** `proxy.OpenViewForExperimentDataItem(std::move(dataitem))` is called.

**Then** the mock records the exact same raw pointer that the `unique_com_ptr` was holding.

| # | Step                                                                         | Expected result             |
|---|------------------------------------------------------------------------------|-----------------------------|
| 1 | Create mock data-item; note its raw address <code>addr</code> .              | <code>addr</code> captured. |
| 2 | Wrap in <code>unique_com_ptr&lt;IDispatch&gt; dataitem</code> .              | Wrapper alive.              |
| 3 | Call <code>proxy.OpenViewForExperimentDataItem(std::move(dataitem))</code> . | View mock records the call. |
| 4 | Assert the pointer captured by the view mock equals <code>addr</code> .      | Passes.                     |

## TC-006 – ExperimentDeviceProxy

Requirement reference: REQ-006.

### TC-006-A: GetName returns correct wide string

**Tags:**

[unit]

**Requirement:**

AC-006-2

**Given** a mock `ExperimentDevice_Dispatch` whose `GetName()` returns a `_bstr_t` with value `L"ECU1"`.

**When** `proxy.GetName()` is called.

**Then** the returned `std::wstring` equals `L"ECU1"`.

| # | Step                                                                            | Expected result                     |
|---|---------------------------------------------------------------------------------|-------------------------------------|
| 1 | Configure mock <code>GetName()</code> to return <code>_bstr_t(L"ECU1")</code> . | Mock ready.                         |
| 2 | Call <code>proxy.GetName()</code> .                                             | Returns <code>std::wstring</code> . |
| 3 | Assert <code>result == L"ECU1"</code> .                                         | Passes.                             |

### TC-006-B: get() returns non-owning raw pointer

**Tags:**

[unit]

**Requirement:**

AC-006-3

**Given** a fully constructed `ExperimentDeviceProxy`.

**When** `proxy.get()` is called.

**Then** the returned pointer equals the internal `p_subject.get()` and the release counter is unaffected.

| # | Step                                                                  | Expected result                                             |
|---|-----------------------------------------------------------------------|-------------------------------------------------------------|
| 1 | Construct proxy from a mock; capture mock address <code>addr</code> . | Proxy alive; counter = 0.                                   |
| 2 | Call <code>proxy.get()</code> .                                       | Returns a pointer equal to <code>addr</code> ; counter = 0. |

## TC-007 – CalibrationScalarDataProxy

Requirement reference: REQ-007.

### TC-007-A: SetImplValue forwards the double argument

**Tags:**

[unit]

**Requirement:**

AC-007-2

**Given** a mock CalibrationScalarData\_Dispatch that records the value passed to SetImplValue.

**When** proxy.SetImplValue(3.14) is called.

**Then** the mock records the value 3.14 and was called exactly once.

| # | Step                                                                  | Expected result                               |
|---|-----------------------------------------------------------------------|-----------------------------------------------|
| 1 | Construct proxy with mock;<br>call counter = 0.                       | Ready.                                        |
| 2 | Call<br>proxy.SetImplValue(3.14).                                     | Mock call counter = 1; recorded value = 3.14. |
| 3 | Assert recorded value ==<br>3.14 (within floating-point<br>equality). | Passes.                                       |

### TC-007-B: ResetValueToRP delegates to COM

**Tags:**

[unit]

**Requirement:**

AC-007-3

**Given** a mock recording calls to ResetValueToRP.

**When** proxy.ResetValueToRP() is called.

**Then** the mock records exactly one call.

| # | Step                                   | Expected result    |
|---|----------------------------------------|--------------------|
| 1 | Construct proxy; reset<br>counter = 0. | Ready.             |
| 2 | Call<br>proxy.ResetValueToRP().        | Reset counter = 1. |

## TC-008 – Experiment

Requirement reference: REQ-008.

For this section all COM proxies are replaced by lightweight mock wrappers that expose the same interface and record calls.

### TC-008-A: add\_param is idempotent

**Tags:**

[unit]

**Requirement:**

AC-008-3

**Given** an Experiment constructed with mock proxies, and a mock that returns a distinct IDispatch object each time GetCalibrationValueInDevice is called.

**When** add\_param("speed") is called twice.

**Then** GetCalibrationValueInDevice is called exactly once (the second call is a no-op).

| # | Step                                             | Expected result                               |
|---|--------------------------------------------------|-----------------------------------------------|
| 1 | Call<br>experiment.add_param("speed").           | COM call count = 1; map size = 1.             |
| 2 | Call<br>experiment.add_param("speed!")<br>again. | COM call count still = 1; map size still = 1. |

### TC-008-B: set\_param throws for unknown parameter

**Tags:**

[unit]

**Requirement:**

AC-008-4

**Given** an Experiment with no parameters added.

**When** set\_param("torque", 50.0) is called.

**Then** std::out\_of\_range is thrown with a message containing "torque".

| # | Step                                                                 | Expected result           |
|---|----------------------------------------------------------------------|---------------------------|
| 1 | Construct experiment with<br>empty parameter map.                    | Ready.                    |
| 2 | Call<br>experiment.set_param("torque",<br>50.0) inside EXPECT_THROW. | std::out_of_range thrown. |
| 3 | Assert message contains<br>"torque".                                 | Passes.                   |

**TC-008-C: set\_param updates registered parameter****Tags:**

[unit]

**Requirement:**

AC-008-4

**Given** an Experiment with "torque" added via add\_param.**When** set\_param("torque", 50.0) is called.**Then** the underlying mock proxy's SetImplValue is called with 50.0.

| # | Step                                       | Expected result                                        |
|---|--------------------------------------------|--------------------------------------------------------|
| 1 | Call experiment.add_param("torque", 50.0). | Mock proxy for "torque" stored internally.             |
| 2 | Call experiment.set_param("torque", 50.0). | SetImplValue(50.0) called once on corresponding proxy. |

**TC-008-D: reset calls ResetValueToRP on all proxies and clears state****Tags:**

[unit]

**Requirement:**

AC-008-5

**Given** an Experiment with two parameters "a" and "b" registered.**When** reset() is called.**Then** ResetValueToRP() is called on both proxies, and subsequent set\_param for either name throws std::out\_of\_range.

| # | Step                      | Expected result                                                |
|---|---------------------------|----------------------------------------------------------------|
| 1 | Add params "a" and "b".   | Both stored; map size = 2.                                     |
| 2 | Call reset().             | ResetValueToRP() called on both mocks; map and vector cleared. |
| 3 | Call set_param("a", 0.0). | std::out_of_range thrown (state cleared).                      |

**TC-008-E: stop\_recording sleeps at least 40 ms before delegating****Tags:**

[unit]

**Requirement:**

AC-008-2, AC-008-7



**Given** an Experiment backed by mocks, and a high-resolution timer.

**When** `stop_recording("")` is called.

**Then** the elapsed wall-clock time from entry to the first COM call is at least 40 ms; `StopRecordingAndSave()` is called before `StopMeasurement()`.

| # | Step                                                                                            | Expected result                   |
|---|-------------------------------------------------------------------------------------------------|-----------------------------------|
| 1 | Record <code>t_start</code> using <code>std::chrono::steady_clock::now()</code> .               | Baseline captured.                |
| 2 | Call <code>experiment.stop_recording("")</code> .                                               | Method blocks for $\geq 40$ ms.   |
| 3 | Record <code>t_end</code> after method returns.                                                 | $t_{end} - t_{start} \geq 40$ ms. |
| 4 | Assert mock call order: <code>StopRecordingAndSave</code> before <code>StopMeasurement</code> . | Order correct.                    |
| 5 | Assert <code>reset()</code> was invoked (map is empty).                                         | Passes.                           |

#### TC-008-F: `start_recording` delegates to COM

**Tags:**

[unit]

**Requirement:**

AC-008-1

**Given** an Experiment backed by a mock with call counter = 0.

**When** `start_recording()` is called.

**Then** `StartRecording()` on the underlying mock is called exactly once.

| # | Step                                             | Expected result                               |
|---|--------------------------------------------------|-----------------------------------------------|
| 1 | Call <code>experiment.start_recording()</code> . | Mock <code>StartRecording</code> counter = 1. |

## TC-009 – Session

Requirement reference: REQ-009.

*Tests TC-009-A through TC-009-C require a Windows host with INCA installed ([integration]). TC-009-D is a compile-time check ([unit]).*

### TC-009-A: Session::connect() succeeds with INCA running

**Tags:**

[integration]

**Requirement:**

AC-009-1, AC-009-3, AC-009-5, AC-009-6

**Given** INCA is installed, running, has an experiment open with at least one device online, and COM has been initialised via CoInitializeEx.

**When** Session::connect() is called.

**Then** a valid Session is returned, and session.experiment() returns a reference to the owned Experiment.

| # | Step                             | Expected result                                |
|---|----------------------------------|------------------------------------------------|
| 1 | Initialise COM (CoInitializeEx). | S_OK or S_FALSE.                               |
| 2 | Call Session::connect().         | No exception; Session object returned.         |
| 3 | Call session.experiment().       | Returns reference without crash.               |
| 4 | Destroy Session.                 | DisconnectFromTool() called on INCA; no crash. |

### TC-009-B: connect() throws when INCA is not installed

**Tags:**

[integration]

**Requirement:**

AC-009-2

**Given** INCA is *not* installed on the test host (or the CLSID is unregistered).

**When** Session::connect() is called.

**Then** std::runtime\_error is thrown with a message containing the word "INCA".

| # | Step                                                                    | Expected result                         |
|---|-------------------------------------------------------------------------|-----------------------------------------|
| 1 | Ensure INCA CLSID is not registered (or run on a machine without INCA). | Environment verified.                   |
| 2 | Call <code>Session::connect()</code> inside try/catch.                  | <code>std::runtime_error</code> caught. |
| 3 | Assert message contains "INCA".                                         | Passes.                                 |

#### TC-009-C: `connect()` throws when no device is online

**Tags:**

[integration]

**Requirement:**

AC-009-4

**Given** INCA is running, an experiment is open, but *no* device is configured or online.

**When** `Session::connect()` is called.

**Then** `std::runtime_error` is thrown with a message indicating no device is available.

| # | Step                                             | Expected result                         |
|---|--------------------------------------------------|-----------------------------------------|
| 1 | Configure INCA experiment with no online device. | Environment ready.                      |
| 2 | Call <code>Session::connect()</code> .           | <code>std::runtime_error</code> thrown. |
| 3 | Assert message mentions device not available.    | Passes.                                 |

#### TC-009-D: Session is non-copyable and non-movable

**Tags:**

[unit] (compile-time check)

**Requirement:**

AC-009-7

**Given** the `Session` type definition.

**When** copy and move traits are evaluated at compile time.

**Then** all four of copy-constructible, copy-assignable, move-constructible, and move-assignable are `false`.

| # | Step   | Expected result                                                            |
|---|--------|----------------------------------------------------------------------------|
| 1 | Assert | Holds.<br><code>!std::is_copy_constructible_v&lt;inca::Session&gt;.</code> |
| 2 | Assert | Holds.<br><code>!std::is_copy_assignable_v&lt;inca::Session&gt;.</code>    |
| 3 | Assert | Holds.<br><code>!std::is_move_constructible_v&lt;inca::Session&gt;.</code> |
| 4 | Assert | Holds.<br><code>!std::is_move_assignable_v&lt;inca::Session&gt;.</code>    |